

Conditional and Independent-Event Probability

Two-way tables, conditional probability, testing independence, and the multiplication rule

Directions:

- Give every probability as an exact fraction in lowest terms unless the problem asks for a decimal.
- Problems 1–4 and 10 use the survey table below. All counts are numbers of students.
- Write the probability statement you are computing (for example $P(A | B)$) before you compute it.

Survey of 120 students at one school (counts of students)

	Plays a sport	No sport	Total
Plays an instrument	18	22	40
Plays no instrument	42	38	80
Total	60	60	120

- 1.** One of the 120 surveyed students is chosen at random. Find the probability that the student plays an instrument.

Answer: _____

- 2.** A student is chosen at random *from those who play an instrument*. Find the probability that this student also plays a sport.

Answer: _____

3. A student is chosen at random from those who play a sport. Find the probability that this student plays an instrument.

Answer: _____

4. Test whether “plays a sport” and “plays an instrument” are independent for these 120 students.

(a) Compute $P(\text{sport}) \cdot P(\text{instrument})$. _____

(b) Compute $P(\text{sport and instrument})$. _____

(c) Explain what the comparison of (a) and (b) tells you about independence.

5. Two cards are dealt from a standard 52-card deck without replacement. Find the probability that both cards are hearts. (There are 13 hearts in the deck.)

Answer: _____

6. A machine produces parts independently, and each part is defective with probability 0.04. Two parts are selected. Find the probability that both are defective.

Answer: _____

7. On the same production line, each part is good with probability 0.96, independently of the others. Five parts are selected. Find the probability that at least one of the five is defective. Round to four decimal places.

Answer: _____

8. A trial is run twice, independently. Each trial succeeds with probability p and fails with probability q , where $p + q = 1$.

(a) Expand $(p + q)^2$. _____

(b) Explain which two-trial outcome each term of your expansion counts, and why one term carries a factor of 2.

9. In a screening programme, 5% of people have a certain condition. The test returns a positive result for 90% of people who have it, and for 8% of people who do not.

- (a) Find the probability that a randomly chosen person tests positive. Round to three decimal places. _____
- (b) Given that a person tested positive, find the probability that they have the condition. Round to three decimal places. _____

10. Return to the survey table of 120 students.

- (a) Find the probability that a randomly chosen student plays neither a sport nor an instrument. _____
- (b) Find the probability that a student plays no instrument, given that the student plays no sport. _____
- (c) Explain whether knowing that a student plays no sport changes the chance that they play no instrument.